

Tanvir Sohail

Postdoctoral Research Associate

National Center for Computational Sciences,
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Professional Appointments

Postdoctoral Research Associate

Sept 2023–Present

Oak Ridge National Laboratory
National Center for Computational Sciences
Advanced Computing for Chemistry and Materials Group
Mentor: Dr. Swarnava Ghosh

Associate Instructor

June 2023–Aug 2023

The University of Alabama, Tuscaloosa
Department of Aerospace and Mechanics Engineering

Instructor of Record

Dec 2021–May 2023

The University of Alabama, Tuscaloosa
Department of Aerospace and Mechanics Engineering
Supervisor: Dr. Mark Barkey

Graduate Research Assistant

Jan 2019–May 2023

The University of Alabama, Tuscaloosa
Department of Aerospace and Mechanics Engineering
Mentor: Dr. Samit Roy

Education

Ph.D. in Aerospace Engineering and Mechanics

May 2021–June 2023

The University of Alabama, Tuscaloosa, AL

M.S. in Aerospace Engineering and Mechanics

Jan 2019–May 2021

The University of Alabama, Tuscaloosa, AL

B.E. in Construction Engineering

Aug 2014–June 2018

Jadavpur University, Kolkata, India

Research Interests

- ★ Multiscale modeling and simulation of materials
- ★ Machine learning and uncertainty quantification
- ★ Quantum materials, skyrmions and topological magnetism
- ★ First-principles and atomistic simulations
- ★ Quantum computing for materials

Technical Proficiencies

- ◆ **Machine Learning:** classification, regression, ensembles, time series, PCA, clustering
- ◆ **Deep Learning:** PyTorch, TensorFlow, CNN, RNN/LSTM/GRU, RNO
- ◆ **HPC/Workflow:** SLURM arrays, MPI/OpenMP, GPUs (ROCm), RADICAL-EnTK
- ◆ **LLM Systems and Distributed Training:** Megatron-LM, DeepSpeed (ZeRO-2/3), PyTorch Distributed, ROCm/RCCL
- ◆ **Atomistic/Ab-Initio:** LAMMPS, ReaxFF, PCFF, VASP, Quantum ESPRESSO, FLEUR
- ◆ **Finite Element:** ABAQUS/Explicit (VUMAT), HyperMesh, SAP2000
- ◆ **Programming:** Python, C/C++, Fortran, MATLAB, R; Qiskit, PennyLane
- ◆ **Tools:** JupyterLab, PyCharm, Git/GitHub

Research Experiences

1. Postdoctoral Research Associate

Sept 2023–Present

Topological Magnetic Defects for Spintronic Memory

- Simulated bloch points, screw dislocations, and skyrmions to quantify defect core strength under magnetic field and strain.
- Identified stability regimes and mapped phase behavior relevant to skyrmion-based memory devices.

Atomistic Design of Composite Interphases

- Performed molecular dynamics (MD) simulations multifunctional composite with controlled interfacial chemistry and crosslinking.
- Established structure–property relations linking interphase ordering to interfacial strength.

Exascale Ensemble Workflows on Frontier

- Designed and executed ensemble workflows for MD sweeps using SLURM arrays and RADICAL-EnTK.
- Benchmarked throughput and fault-tolerance, demonstrating robust scaling on OakRidge Leadership Computing Facility (OLCF) Frontier.

Strain Engineering of Quantum Materials

- Applied first-principles calculations to molybdenum disulfide (MoS_2) monolayers and yttrium-doped manganese bismuth telluride ($MnBi_2Te_4$, MBT) under strain.
- Quantified band topology changes, electronic band-gap shifts, and anisotropy, providing design insights for tunable quantum devices.

Thermal Transport at Polymer–CNT Interfaces

- Conducted equilibrium and non-equilibrium MD to design covalent linkers at CNT–polymer interfaces.
- Demonstrated how linker length/chemistry modulate Kapitza resistance and composite conductivity.

Multiscale Surrogate Modeling of Polyurea Ballistic Response

- Developed RNO surrogates trained on temperature- and strain-rate-dependent MD simulations of polyurea; incorporated Optimal Uncertainty Quantification to capture variability arising from MD temperature effects.
- Established a multiscale framework where the RNO surrogate emulates finite element (FEM/VUMAT) constitutive models, achieving orders-of-magnitude speedup while retaining predictive fidelity for stress–strain histories from MD.

Defect Thermodynamics in High-Entropy Alloys

- Performed large-scale MD simulations of Ni–Cr–Co–Fe and Ni–Cr–Co alloys to quantify vacancy formation energies under applied strain and varying local chemical environments.
- Developed graph neural network surrogates trained on MD data to predict vacancy formation energies directly from atomic neighborhood descriptors, enabling rapid exploration of defect energetics in chemically complex alloys.

Quantum Algorithms for Strongly Correlated Materials

- Implemented the Variational Quantum Eigensolver and Variational Quantum Deflation algorithms in Qiskit and PennyLane to solve strongly correlated model Hamiltonians relevant to quantum materials.
- Prototyped quantum Dynamical Mean-Field Theory impurity solvers with Message Passing Interface based parameter sweeps, validating results against exact diagonalization benchmarks.

Pressure-Dependent Plasticity in Metals

- Modeled the influence of hydrostatic pressure on stacking-fault energy (SFE) and Peierls stress in Cu and tantalum (Ta) using atomistic simulations.
- Established quantitative links between pressure-dependent SFE ($\gamma_{SF}(p)$) and Peierls barrier variations with predicted changes in dislocation mobility and slip resistance.

High-Performance Computing Code Development on Frontier

- Ported the lattice kinetic Monte Carlo library to GPUs on OLCF Frontier, achieving significant speedups through GPU-aware MPI and HIP optimization for large-scale replica simulations.
- Built and optimized multiple HPC materials science codes on Frontier, including finite element framework FEniCS, LAMMPS, VASP, SPARC, and FLEUR.

Large-Scale LLM Training on Frontier

- Deployed distributed training of large language models using Megatron-LM and DeepSpeed on OLCF Frontier (AMD MI250X, ROCm/RCCL), combining data-, tensor-, and pipeline-parallelism with ZeRO optimizer (Stages 2/3), gradient checkpointing, and mixed precision (FP16/BF16) for scalable throughput.
- Engineered end-to-end training workflows: SLURM + PyTorch Distributed launchers, efficient checkpointing/restart for preemption tolerance, tokenization and dataset curation (BPE/SentencePiece, dedup/filtering), and evaluation via perplexity and zero-/few-shot probes to track convergence and quality.

2. Graduate Research Assistant

Jan 2019–May 2023

The University of Alabama, Tuscaloosa, AL

Atomistic J-Integral and Nanoscale Fracture in Amorphous Polymers

- Built EPON-862 models (PCFF) and performed fracture simulations with ReaxFF to quantify atomistic J including entropic contributions.
- Established flaw-size regimes where LEFM holds at the nanoscale and verified a crack-size-independent cohesive traction-separation law.

Interfacial Mechanics in CNT/Polymer Systems

- Simulated PAN and ABS matrices around CNTs with controlled crosslink density; quantified polymer-CNT affinity via interaction energy.
- Modeled carbon-fiber surface asperities from AFM via Fourier series; predicted IFSS gains from MWNT-polymer-asperity interactions.

Coal Nanostructure and Bioconversion Stresses

- Built bituminous coal models and mapped virial stresses from MD to continuum measures.
- Showed internal gas-generation pressures markedly alter effective response in confined vs. unconfined cases.

Chaotic Dynamics Forecasting

- Predicted Chua-circuit trajectories with RNN/LSTM/GRU under one-step, multi-step, and sequence forecasting.
- LSTM with sequence forecasting best preserved ergodic features; hyperparameters optimized with GridSearchCV.

Large-Scale Data: LHC and Spin Systems

- Performed EDA on LHC data (incl. K-means outlier detection); trained BDT and DNN for Higgs/background (28 features, 11M samples), achieving AUC 0.81 (BDT) and 0.885 (DNN).
- Predicted spin-interaction variables from magnetization curves using ML; Gaussian Process Regression yielded lowest MSE of 0.12.

Peer-Reviewed Journal Articles

1. **Tanvir Sohail**, Burigede Liu, Swarnava Ghosh, “Neural operator accelerated atomistic to continuum concurrent multiscale simulations of viscoelasticity,”
<https://arxiv.org/pdf/2603.27430>

2. Sumit Gupta, **Tanvir Sohail**, Marti Checa, Michael D. Toomey, Logan T. Kearney, Rajni Chahal, Sargun S. Rohewal, Nihal Kanbargi, Swarnava Ghosh, Liam Collins, David McConnell, Ilia N. Ivanov, Amit K. Naskar, and Christopher C. Bowland, “Designing Physicochemically-Ordered Interphases for High-Performance Composites,” *Advanced Functional Materials* 35.19 (2025): 2502972.
3. Antigoni Georgiadou, Henry Monge-Camacho, **Tanvir Sohail**, Swarnava Ghosh, Arjun Valiya Parambathu, Dilipkumar N. Asthagiri, Dmytro Bykov, Tushar Athawale, and Thomas L. Beck, “Ensemble Simulations on Leadership Computing Systems,” in *SC24-W: Workshops of the International Conference for High Performance Computing, Networking, Storage and Analysis*, pp. 394–401, IEEE, 2024.
4. Sumit Gupta, **Tanvir Sohail**, Marti Checa, Sargun Rohewal, Micheal Toomey, Nihal Kanbargi, Josh Damron, Liam Collins, Logan Kearney, Amit Naskar, and Christopher Bowland, “Enhancing composite toughness through hierarchical interphase formation,” *Advanced Science* 11.6 (2024): 2305642.
5. Sankha Subhra Aditya, **Tanvir Sohail** and Samit Roy, ”An atomistic-continuum concurrent statistical coupling technique for amorphous materials using anchor points,” *Modelling and Simulation in Materials Science and Engineering* 31.7 (2023): 075012.
6. **Tanvir Sohail** and Samit Roy, “Application of the atomistic J-integral including entropic effect to study the validity of LEFM at the nanoscale in amorphous materials,” *Theoretical and Applied Fracture Mechanics* 125 (2023): 103913.
7. **Tanvir Sohail**, Rebekah Sweat, Ray Baughman, Hongbing Lu, and Samit Roy, “Prediction of interfacial shear strength of CNT overwrapped carbon fibers using molecular dynamics and Fourier series decomposition of surface asperities,” *Composite Interfaces* 30.9 (2023): 1011-1034.
8. **Tanvir Sohail**, Sushan Nakarmi, Rebekah Sweat, Ray Baughman, Hongbing Lu, and Samit Roy, “Simulation of surface asperities on a carbon fiber using molecular dynamics and Fourier series decomposition to predict interfacial shear strength in polymer matrix composites,” *Composite Interfaces* 29.8 (2022): 947-970.
9. Sushan Nakarmi, **Tanvir Sohail** and Samit Roy, “An investigation of flaw-size independence of cohesive traction-separation law at the nanoscale: a molecular dynamics study,” *Engineering Fracture Mechanics* 215 (2019): 235–245.
10. Sumit Saha, **Tanvir Sohail**, Arijit Bose, and Jafar Sadak Ali, “Damage identification of structures using limited power mode shapes and power modal curvature,” in *Advances in Rotor Dynamics, Control and Structural Health Monitoring* (2020): 257–274.

Peer-Reviewed Conference Papers

1. Sumit Gupta, **Tanvir Sohail**, Amit K. Naskar, and Christopher C. Bowland, “Multiscale modeling-enabled design of multifunctional composites.” *Nondestructive Characterization and Monitoring of Advanced Materials, Aerospace, Civil Infrastructure, and Transportation XVIII*, vol. 12950, pp. 204–213. SPIE, 2024.
2. Samit Roy, and **Tanvir Sohail**, “Investigating the influence of nanoparticle size and orientation on failure in notched epoxy nanocomposites using the atomistic J-integral.”

ASC 37th Technical Conference, 2022

3. Samit Roy, and **Tanvir Sohail**, “Failure simulation in notched polymer nanocomposites using molecular dynamics.” *AIAA SciTech, 2022*
4. Samit Roy, and **Tanvir Sohail**, “Fracture simulation in polymer nanocomposites.” *ASC 36th Technical Conference, 2021*
5. Samit Roy, and **Tanvir Sohail**, “Fracture behavior in polymer nanocomposites: a molecular dynamics study.” *AIAA SciTech, 2021*
6. Samit Roy, **Tanvir Sohail**, and Jony Bodiuzzaman, “Modeling the effect of nanoparticle size on fracture toughness of nanocomposites.” *AIAA SciTech, 2020*
7. Jony Bodiuzzaman, **Tanvir Sohail**, and Samit Roy, “Improving fracture toughness at the nanoscale using length-scale effect.” *ASC 34th Technical Conference, 2019*
8. **Tanvir Sohail**, Sumit Saha, and Jafar Sadak Ali, “Experimentation and numerical study of magnetorheological damper using Bou–Wen model.” *ICTACEM, 2017*
9. **Tanvir Sohail**, Sumit Sah, Debasish Bandhyopadhyay, “Parameter identification of frame structure by noisy limited static strain data.” *ICAM, 2017*

Presentations and Invited Talks

1. “Multiscale Modeling of History-Dependent Materials Using Continuous -Time Neural Operators,” APS Global Summit , Denver, CO, Mar 16th, 2026.
2. “RNO: A Continuous-Time Neural Operator for Efficient Multiscale Constitutive Modeling,” Poster, AI Expo, ORNL, Apr 2, 2025.
3. “Skyrmions: Tunable Magnetic Structures for Next-Generation Spintronics,” APS Global Summit, Anaheim, CA, Mar 18, 2025.
4. “Strain-Tuned Skyrmions — Revolutionizing Next Generation Spintronics and Memory,” MRS Fall Meeting and Exhibition, Boston, MA, Dec 2, 2024.
5. “Atomistic Modeling-enabled Interface Design for Tougher Composites,” ASME SMASIS 2024, Atlanta, GA, Sept 10, 2024.
6. “Tuning Skyrmion Physics for Next-Generation Spintronics,” 12th ORNL Postdoctoral Research Associate Organization Symposium, July 11, 2024.
7. “Fracture Mechanics Redefined: A Multiscale Approach to Polymer Nanocomposites,” National Center for Computational Sciences, ORNL, July 11, 2023 (invited).
8. “Computational Study of Fracture in Polymer Nanocomposite,” Chemical and Biomolecular Engineering, University of Delaware, May 25, 2023 (invited).
9. “Multiscale Modeling of Fracture in Polymer Nanocomposites,” Mechanical, Aerospace and Nuclear Engineering, RPI, May 6, 2023 (invited).

Awards and Honors

- NSF Future Faculty Travel Award, SES 2025, Atlanta, GA
- Selected Participant, Argonne Training Program on Extreme-Scale Computing (ATPESC 2025)
- ASME SMASIS 2024 “Best Paper in Mechanics and Material Systems Award”
- Finalist, ORNL “Your Science in a Nutshell” Competition (2024)
- Distinguished Postdoctoral Fellowship, ORNL Leadership Computing Facility
- Outstanding Student Instruction Award, University of Alabama (2021, 2022)

Synergistic Activities and Service

- * **Session Chair:** *Recent Advances in Computational Methods and Simulation for Fusion Materials* (APS Global Summit, Anaheim, CA, Mar 17, 2025); *Optical SHM Technologies* (ASME SMASIS 2024, Atlanta, GA, Sept 10, 2024)
- * **Proposal Reviewer:** INCITE; SummitPlus (2023–present)
- * **Journal Reviewer:** Scientific Reports; Progress in Additive Manufacturing; Journal of Reinforced Plastics and Composites; MRS Advances
- * **Judge:** ORNL GEM Fellows Tech Talks (July 23, 2024); Undergraduate Research and Creative Activity Conference, University of Alabama (Mar 29, 2022)
- * **Organizing Committee:** 12th ORPA (ORNL Postdoctoral) Research Symposium, July 14–15, 2024
- * **Workshops and Training:** OpenMP Training Series, NERSC (May–Oct 2024); ORNL Powerful Proposal Writing
- * **Code Contributions:** GPU porting of operator-learning workflows on Frontier; GPU-accelerated Kinetic Monte Carlo (prototype) on Frontier
- * **Quantum Computing code** Wrote and compiled codes on IBM machines.

Grants and Proposals

- ◆ **PI:** Elucidating Strain-Dependent Vacancy Formation Energies in High-Entropy Alloys Through Computational Analysis (ACCESS - awarded), 750,000 ACCESS credits.
- ◆ **Co-PI:** In silico atomistic design of hierarchically nanostructured interphases for high-performance composites (OLCF Director’s Discretion - awarded), 20,000 node hours.
- ◆ **Co-PI:** Computational Design of Architected Interphases for High-Performance Mechano-Phonoelectronic Materials (OLCF Director’s Discretion - awarded), 20,000 node hours.
- ◆ **Co-PI:** Deep Mechanician: Automated Generative AI for Constitutive Modeling (OLCF Director’s Discretion - awarded), 20,000 node hours.

References

- ✉ Dr. Swarnava Ghosh, Computational Scientist,
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- ✉ Dr. Sumit Gupta, R&D Staff Member,

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✉ Dr. Samit. Roy, William D. Jordan Professor,
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